

UNIQUENESS OF THE 't HOOFT-POLYAKOV MAGNETIC MONOPOLE SOLUTION

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In the $O(3)$ gauge theory with a triplet of scalars, we show the uniqueness of the Wu–Yang–'t Hooft–Polyakov–Julia–Zee solution under the assumptions: 1) static, finite energy solution, 2) separability of radial and angular variables.

Topological arguments [1] allow the existence in the $SU(2)$ gauge theory with a triplet of Higgs scalars of magnetic monopoles of strength $M/2$ where M is an integer. 't Hooft [2] and Polyakov [3] constructed a solution of the equations of motion with $M = 1$, and Julia and Zee [4] generalized it to the case where the magnetic monopole has also an electric charge (dyon). We wish to show that no solutions with $M > 1$ exist under a simple set of assumptions.

The Lagrangian density for the model is:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^a F^{\mu\nu a} + \frac{1}{2}\Pi_{\mu a}\Pi_{\mu a} - \frac{\lambda}{4}[Q^a Q^a - \mu^2/\lambda]^2 \quad (1)$$

where $a = 1, 2, 3$, and

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + e\epsilon^{abc}A_\mu^b A_\nu^c \quad (2)$$

$$\Pi_\mu^a = \partial_\mu Q^a + e\epsilon^{abc}A_\mu^b Q^c. \quad (3)$$

We shall look for solutions of the equations of motion satisfying the following assumptions:

1. Time-independence

2. Separability into radial (r) and angular (θ, ϕ) coordinates, i.e.:

$$Q^a = f(r)\Omega^a(\theta, \phi) \quad (4)$$

$$A_i^a = \frac{1}{e} \frac{h(r)}{r} B_i^a(\theta, \phi), \quad (5)$$

$$A_0^a = k(r)C^a(\theta, \phi), \quad (6)$$

where all these functions are a priori unknown (we do *not* assume the specific ansatz of Wu–Yang [5] 't Hooft–Polyakov–Julia–Zee, but we shall derive it). We assume analyticity in the variables r, θ, ϕ and single valuedness.

3. The solutions have finite energy.

The energy of the solution is given by $\int d^3x T_{00}$ where $T_{\mu\nu}$ is the covariant energy-momentum tensor coupled to the gravitational field. T_{00} is given by

$$T_{00} = F_{0i}^a F_{0i}^a + \Pi_0^a \Pi_0^a - \mathcal{L} \quad (7)$$

and is easily seen to be a sum of positive quantities. Therefore, each of them must be integrable and go to zero fast enough at infinity for the energy to be finite.

First the potential must vanish at infinity. Therefore if

$$Q^a \underset{r \rightarrow \infty}{=} f_0 \Omega^a(\theta, \phi) \quad (8)$$

one has

$$\mu^2/\lambda = f_0^2 \Omega^a \Omega^a. \quad (9)$$

So $\Omega^a \Omega^a$ is a constant which can be normalized to one

$$\Omega^a \Omega^a = 1 \quad (10)$$

Next $(\Pi_0^a)^2$ must be integrable, so one needs Π_0^a to go to zero faster than r^{-1} at infinity. Since $\Pi_0 = ekfC \times \Omega$ and since f goes to a constant (non zero magnetic charge), and since k behaves as r^{-1} (non zero electric charge), it requires C and Ω to be parallel. So, one normalizes C such that $rk(r) \underset{r \rightarrow \infty}{\rightarrow} 1$

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$$A_0 = k(r) \Omega a(\theta, \phi). \quad (11)$$

We shall not discuss intermediate cases where k goes down faster than r^{-1} without being zero, for instance when the electric charge distribution is a dipole.

Finally the $(\Pi_i^a)^2$ term must be integrable. So Π_i^a must vanish at infinity. This, together with the factorizability condition is seen to imply:

$$A_i = \frac{1}{e} h(r) [\Omega \times \partial_i \Omega + (B_i/r) \Omega] \quad (12)$$

where B_i are functions of the angular variables only, and

$$\lim_{r \rightarrow \infty} h(r) = -1.$$

Now we write down the equation of motion of the Higgs fields. It depends only upon the unknown functions f, h, Ω, B_i and turns out to be

$$L^2 \Omega + h(r) \Omega (\Omega \cdot L^2 \Omega) - q(r) \Omega + rh(\Omega \times \partial_i \Omega) B_i = 0 \quad (13)$$

where

$$q(r) = f^{-1}(1+h)^{-1} [(r^2 f')' + \mu^2 r^2 f - \lambda r^2 f^3] \quad (14)$$

and L^2 is the orbital angular momentum operator

$$L^2 = -\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) - \frac{1}{\sin^2 \theta} \partial_\phi^2. \quad (15)$$

Taking the scalar product of this equation by Ω yields:

$$\Omega \cdot L^2 \Omega = \frac{q(r)}{1+h(r)} = c. \quad (16)$$

So that we obtain the equation:

$$L^2 \Omega - c \Omega + rh(\Omega \times \partial_i \Omega) B_i = 0. \quad (17)$$

If $B_i \neq 0$ this equation implies that h is strictly a constant, i.e. -1 .

Then the $(F_{ij})^2$ term behaves as r^{-4} at the origin, and therefore the solution does not have finite energy. So we set $B_i = 0$. It implies from eq. (12) that the radial component of A_μ^i vanishes. Eq. (16) now implies that the Ω are eigenvectors of L^2 with a common eigenvalue c

According to the hypothesis of analyticity and single valuedness on a sphere, c must be of the form $l(l+1)$, where l is an integer. Therefore the equations

are now separated into radial and orbital parts

$$L^2 \Omega = l(l+1) \Omega \quad (18)$$

$$q(r) = l(l+1)(1+h). \quad (19)$$

Eq. (19) is easily seen to reduce to the 't Hooft-Polyakov equation in the $l=1$ case.

Let us now show, to complete the proof, that $l=0, 1$ are the only allowed cases. This is a consequence of the following general theorem, applied in the case $N=3$.

Th. : Let there be N real analytic, single valued functions on the sphere $\Omega^a(\theta, \phi)$ satisfying:

$$a) \Omega^a \Omega^a = 1 \quad (20)$$

and

$$b) L^2 \Omega^a = l(l+1) \Omega^a. \quad (21)$$

Then the only allowed values for l are such that:

$$2l+1 \leq N.$$

Proof: From b) and the assumption of analyticity and single-valuedness it follows that

$$\Omega^a(\theta, \phi) = \sum_{-l \leq m \leq l} Y_{l,m}(\theta, \phi) C_{m,a}, \quad (22)$$

where the $C_{m,a}$ are coefficients satisfying the conditions:

$$(-1)^m C_{-m,a}^* = C_{m,a} \quad (23)$$

and the $Y_{l,m}$ are the spherical harmonics.

We now use the non linear constraint a). Using the addition theorem of spherical harmonics, and their independence, one sees that it implies:

$$\sum_{m,m',a} \left[\frac{(2l+1)^2}{(4\pi)^2(2J+1)} \right]^{1/2} \times \langle ll00|J0 \rangle \langle llmm'|JM \rangle C_{m,a} C_{m'a} = \delta_{J0} \delta_{M0} \quad (24)$$

for all $J \geq 0$ integer, $-J \leq M \leq J$, M integer. We use the notations and conventions of [6] for the Clebsch-Gordan coefficients. $\langle ll00|J0 \rangle$ vanishes for odd J , and otherwise it is nonzero. Therefore, we deduce the equation:

$$\sum_{m,m',a} \langle llmm'|JM \rangle C_{ma} C_{m'a} = \frac{4\pi}{(2l+1)^{1/2}} (-1)^l \delta_{J,0} \delta_{M,0} \quad (25)$$

valid for all even J .

In order to be able to sum on all J, M and use the orthogonality properties of the Clebsch-Gordan coefficients, we multiply this equation by

$$\frac{1}{2} [\langle llm_1m_2 | JM \rangle + \langle llm_2m_1 | JM \rangle]$$

which vanishes for odd J , and sum over all J, M . One obtains the relation:

$$\sum_a C_{m_1,a} C_{m_2,a} = \frac{4\pi}{2l+1} (-1)^{m_1} \delta_{m_1+m_2,0}. \quad (26)$$

This can be considered as a matrix equation. The right hand side matrix is a $(2l+1) \times (2l+1)$ matrix, of determinant non-zero. Therefore its rank is $2l+1$. The left hand side is a $(2l+1) \times (2l+1)$ matrix written as the product of a $N \times (2l+1)$ matrix by a $(2l+1) \times N$ matrix. The rank of each of these is smaller than or equal to $\min(N, 2l+1)$. As the rank of the product of matrices is smaller than or equal to the rank of each of the matrices of the product, we get the inequality:

$$2l+1 \leq \min(N, 2l+1). \quad (27)$$

This inequality can be satisfied only if $2l+1 \leq N$ hence the theorem.

So in the case of an isotriplet of fields ($N=3$) the only allowed values for l are 0 and 1. $l=0$ is just a constant solution in θ, ϕ and has zero magnetic monopole charge. $l=1$ reproduces the usual ansatz, by choosing the axis in isospin space in a convenient fashion. Indeed one has

$$\Omega = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$

$$= \left(\frac{4\pi}{3} \right)^{1/2} \left[-\frac{Y_{1,1} + Y_{1,-1}}{\sqrt{2}}; \frac{-Y_{1,1} - Y_{1,-1}}{\sqrt{2}}; Y_{1,0} \right].$$

Then by exploiting the equations of motion of the gauge fields one obtains the other usual radial equations which have been shown to admit a unique solution in the case $\lambda=0$ [7], and the result $a(\theta, \phi)$ in A_0 is a constant.

This proves the uniqueness of the 't Hooft—Polyakov—Julia—Zee ansatz, and the inexistence (under the conditions stated in the abstract) of monopoles of higher strength in this model.

During the writing of this paper, we learnt that L. Michel, O' Raifeartaigh and K.C. Wali have obtained a similar result under somewhat more general hypothesis, and in particular for larger gauge groups. We thank Prof. O' Raifeartaigh for communicating us these results. One of us (F.S.) wishes to thank the Laboratoire de Physique Théorique de l'Ecole Normale Supérieure for its hospitality.

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